Pizza Sales Analysis

SQL Data Insights

Comprehensive breakdown of orders, revenue, and trends.

Data Overview & Structure

Tables Used

- **Orders:** Order IDs, timestamps.
- **Order_Details:** Quantities, Pizza IDs.
- **Pizza:** Prices, Sizes, Types.
- **Pizza_Types:** Names, Categories, Ingredients.

Basic Performance Metrics

1. Total Orders

```
SELECT
    COUNT(ORDER_ID) AS TOTAL_ORDERS
FROM
    ORDERS;
```

2. Total Revenue

```
SELECT
    ROUND(SUM(ORDER_DETAILS.QUANTITY *
PIZZA.PRICE), 2)
    AS TOTAL_SALES
FROM
    ORDER_DETAILS
JOIN PIZZA
    ON PIZZA.PIZZA_ID = ORDER_DETAILS.PIZZA_ID;
```

Product Insights

3. Highest Priced Pizza

```
SELECT PIZZA_TYPES.NAME, PIZZA.PRICE
FROM PIZZA_TYPES
JOIN PIZZA ON PIZZA_TYPES.PIZZA_TYPE_ID =
PIZZA.PIZZA_TYPE_ID
ORDER BY PRICE DESC LIMIT 1;
```

4. Most Common Size

```
SELECT PIZZA.SIZE,
    COUNT(ORDER_DETAILS.ORDER_DETAILS_ID) AS
ORDER_COUNT
FROM PIZZA
JOIN ORDER_DETAILS ON PIZZA.PIZZA_ID =
ORDER_DETAILS.PIZZA_ID
GROUP BY PIZZA.SIZE
ORDER BY ORDER_COUNT DESC;
```

Top 5 Best Sellers

5. Most Ordered Pizza Types

```
SELECT PIZZA_TYPES.NAME,
    SUM(ORDER_DETAILS.QUANTITY) AS QUANTITY
FROM PIZZA_TYPES
JOIN PIZZA
    ON PIZZA_TYPES.PIZZA_TYPE_ID =
PIZZA.PIZZA_TYPE_ID
JOIN ORDER_DETAILS
    ON ORDER_DETAILS.PIZZA_ID = PIZZA.PIZZA_ID
GROUP BY PIZZA_TYPES.NAME
ORDER BY QUANTITY DESC
LIMIT 5;
```

Category Analysis

6. Quantity by Category

```
SELECT PIZZA_TYPES.CATEGORY,
    SUM(ORDER_DETAILS.QUANTITY) AS QUANTITY
FROM PIZZA_TYPES
JOIN PIZZA ON ...
JOIN ORDER_DETAILS ON ...
GROUP BY PIZZA_TYPES.CATEGORY
ORDER BY QUANTITY DESC;
```

7. Pizzas per Category

```
SELECT CATEGORY, COUNT(NAME)
FROM PIZZA_TYPES
GROUP BY CATEGORY;
```

Temporal Trends

8. Hourly Order Distribution

```
SELECT EXTRACT(HOUR FROM ORDER_TIME) AS HOURS,
    COUNT(ORDER_ID) AS ORDERS
FROM ORDERS
GROUP BY HOURS ORDER BY HOURS;
```

9. Avg Pizzas Per Day

```
SELECT ROUND(AVG(QUANTITY), 0)
FROM (
    SELECT ORDERS.ORDER_DATE,
        SUM(ORDER_DETAILS.QUANTITY) AS QUANTITY
    FROM ORDERS
    JOIN ORDER_DETAILS ON ...
    GROUP BY ORDERS.ORDER_DATE
) AS ORDER_QUANTITY;
```

Top Revenue Generators

10. Top 3 Pizzas by Revenue

```
SELECT PIZZA_TYPES.NAME,
    SUM(ORDER_DETAILS.QUANTITY * PIZZA.PRICE)
AS REVENUE
FROM PIZZA_TYPES
JOIN PIZZA
    ON PIZZA_TYPES.PIZZA_TYPE_ID =
PIZZA.PIZZA_TYPE_ID
JOIN ORDER_DETAILS
    ON ORDER_DETAILS.PIZZA_ID = PIZZA.PIZZA_ID
GROUP BY PIZZA_TYPES.NAME
ORDER BY REVENUE DESC
LIMIT 3;
```

Revenue Share by Category

11. Percentage Contribution

```sql
SELECT PIZZA_TYPES.CATEGORY,
    ROUND(
        SUM(ORDER_DETAILS.QUANTITY * PIZZA.PRICE) / (
            SELECT ROUND(SUM(ORDER_DETAILS.QUANTITY * PIZZA.PRICE), 2)
            FROM ORDER_DETAILS
            JOIN PIZZA ON PIZZA.PIZZA_ID = ORDER_DETAILS.PIZZA_ID
        ) * 100, 2
    ) AS REVENUE
FROM PIZZA_TYPES
JOIN PIZZA ON PIZZA_TYPES.PIZZA_TYPE_ID = PIZZA.PIZZA_TYPE_ID
JOIN ORDER_DETAILS ON ORDER_DETAILS.PIZZA_ID = PIZZA.PIZZA_ID
GROUP BY PIZZA_TYPES.CATEGORY
ORDER BY REVENUE DESC;
```

Growth Analysis

12. Cumulative Revenue Over Time

```sql
SELECT ORDER_DATE, REVENUE,
    SUM(REVENUE) OVER (ORDER BY ORDER_DATE) AS CUM_REVENUE
FROM (
    SELECT ORDERS.ORDER_DATE,
        SUM(ORDER_DETAILS.QUANTITY * PIZZA.PRICE) AS REVENUE
    FROM ORDER_DETAILS
    JOIN PIZZA ON ORDER_DETAILS.PIZZA_ID = PIZZA.PIZZA_ID
    JOIN ORDERS ON ORDERS.ORDER_ID = ORDER_DETAILS.ORDER_ID
    GROUP BY ORDERS.ORDER_DATE
) AS SALES
ORDER BY ORDER_DATE;
```

Category Leaderboard

13. Top 3 Revenue Pizzas Per Category

```sql
SELECT NAME, REVENUE FROM (
    SELECT CATEGORY, NAME, REVENUE,
    RANK() OVER(PARTITION BY CATEGORY ORDER BY REVENUE DESC) AS RN
    FROM (
        SELECT PIZZA_TYPES.CATEGORY, PIZZA_TYPES.NAME,
        SUM((ORDER_DETAILS.QUANTITY) * PIZZA.PRICE) AS REVENUE
        FROM PIZZA_TYPES
        JOIN PIZZA ON PIZZA_TYPES.PIZZA_TYPE_ID = PIZZA.PIZZA_TYPE_ID
        JOIN ORDER_DETAILS ON ORDER_DETAILS.PIZZA_ID = PIZZA.PIZZA_ID
        GROUP BY PIZZA_TYPES.CATEGORY, PIZZA_TYPES.NAME
    ) AS A
) AS B
WHERE RN <= 3;
```

Thank You!

SQL Pizza sales Data Analysis

~Vanshika Dubey